HepatoTwin: Meeting 9 Progress Report

Meeting 9, Friday 23 October 2026 · Manojna

Covers the meeting 9 tasks of the weekly action plan (Task 1: twin state and evidence; Task 2: optimiser and recipes). Proposal section numbers are given in brackets.

1. Summary

- **Priorities are now personal (§17.1).** Each patient's 24-hour dietary recall raises or lowers their nutritional priorities. Across 300 held-out test patients the number of distinct priority profiles rose from 11 to 37, and distinct top-10 recipe lists from 16 to 28.
- **A ranking flaw was found and fixed.** Once a recipe met every health reference, the patient's priority levels no longer affected its rank. With the fix, priorities count inside the health gate; the headline fell from 0.899 to 0.802, and adding acceptability (weight 0.1, now default) brings it to 0.831.
- **The scenario engine (Layer 5, §8) works end to end.** It compares a maximum-health scenario with one closer to the patient's usual diet, giving an evidence-graded direction, never a predicted number.
- **Every seven-day meal plan is complete and meets every constraint (§17.6).** 58% use only the patient's own cuisine, 81% once a favourite may repeat, and the rest borrow about one meal from the most similar cuisine.
- **Six evidence rows were added (§17.13)**, bringing the table to 23; observational coffee findings carry their own label and are never used as effects.
- **Indian regional cuisines moved to future work.** The RecipeDB API holds the same 20,000 recipes already used (19 Indian), and the patient cohort is the US population, so US cuisines are the focus.

2. Plan Versus Delivery

Table 1. Meeting 9 tasks from the weekly action plan.

Task	Planned item	Status	Where
1	Integration checkpoint (end-to-end pipeline and demo)	Met earlier; demo now runs Layers 2–5	src/demo.py
1	Layer 5 scenario engine: Recipe A vs B, evidence-graded direction (§8)	Done	scenario_engine.py

Task	Planned item	Status	Where
1	Habitual diet from NHANES 24-hour recall (§17.1)	Done	habitual_diet.py
1	New evidence rows: coffee, fasting, null findings (§17.13)	Done (6 rows)	evidence CSV
2	Diversity re-ranking and 7-day meal plan (§17.6)	Done	meal_plan.py
2	Full RecipeDB including Indian regional cuisines (§17.12)	Moved to future work	Section 7
2	Rating-based acceptability (§17.12)	Done; now in the default ranking	multi_criteria.py
2	Start recruiting 5–10 pilot evaluators	Not started	Section 10
–	Not planned: priority levels ignored inside the health gate	Found and fixed	Section 3

All code is in the project repository; 83 automated tests pass.

3. Personalised Priorities from the 24-Hour Recall (§17.1)

Data. Day-1 nutrient totals (DR1TOT) were taken for both FibroScan cycles, and the USDA Food Patterns Equivalents Database (FPED) day-1 totals for 2017–March 2020, which add added sugars and the food groups needed for the Healthy Eating Index-2020 (HEI-2020). USDA has not released FPED for 2021–2023, so added sugars and HEI-2020 are missing for that cycle. Of 15,300 cohort participants, 9,391 have a recall that is complete, reported as a usual day, and of plausible energy (500–3,500 kcal for women, 800–4,000 for men); only these are used.

Table 2. Median intake among participants with a reliable recall.

Measure	Reference used for recipes	2017–2020	2021–2023
Saturated fat (% of energy)	≤ 10	11.3	11.7
Added sugars (% of energy)	≤ 10	10.4	not available
Total sugars (% of energy)	–	18.8	18.2
Fibre (g per 1,000 kcal)	≥ 14	7.1	7.8
HEI-2020 (0–100)	–	48.5	not available

65% of reliable recalls exceed the saturated-fat reference, 52% the added-sugar reference, and 89% fall short of the fibre reference. HEI-2020 follows the NCI simple scoring algorithm.

Method. Each priority moves one level up when the patient's own intake misses the same reference value used to score recipes, and one level down when it already meets it; without a reliable recall it is unchanged. A single recall is noisy, so it never moves a priority by more than one level and never acts as a diagnosis. The reason is shown with every priority, for example for the worked patient (SEQN 110006): “Excess weight, where surplus energy from sugar matters most; raised: added sugars 23.5% of energy in the 24-hour recall misses the reference ≤ 10”.

Ranking change needed for this to matter. The default ranking puts recipes that meet every reference first. Inside that group every health score was capped at 100, so priority levels

changed nothing: without any recall, 300 patients received only 8 distinct top-10 lists. Inside the gate, half of the health score now measures how far a recipe goes beyond the saturated-fat, sugar and fibre references, weighted by the patient's priority levels (energy is excluded: a much smaller meal is not better for that reason). Figure 1 shows the trade-off; a share of 0.5 was chosen because it quadruples the variety of recommendations while keeping the gated ranking above weighted scoring.

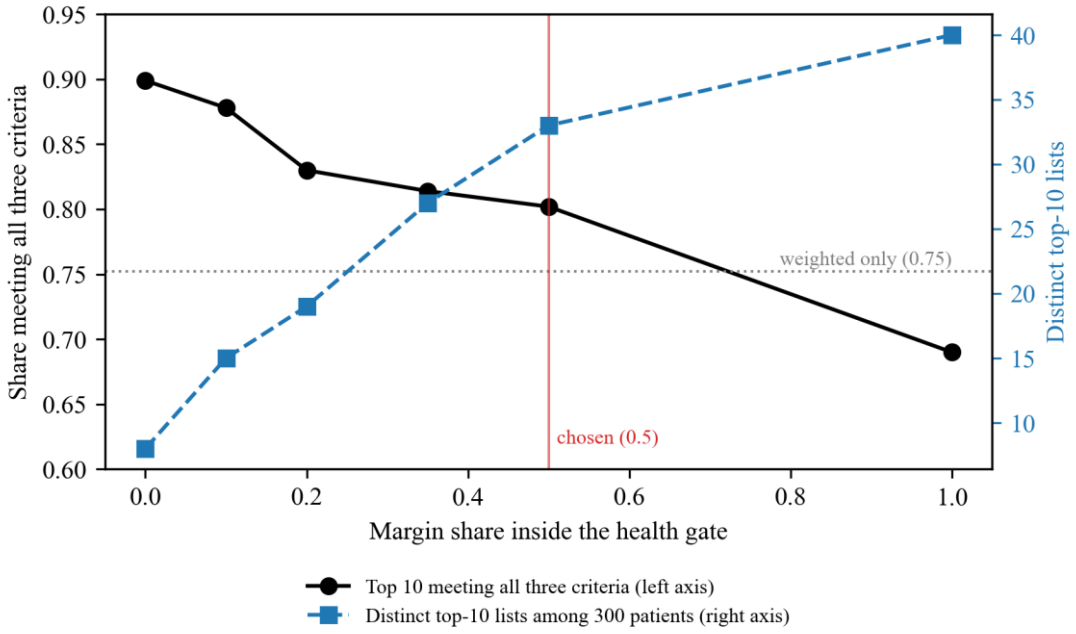


Figure 1. Effect of the margin share on practicality and on how much recommendations differ between patients (300 held-out patients, 24-hour recall used, before acceptability was added to the default).

Table 3. Personalisation with the same engine, without and with the 24-hour recall.

Measure	Without recall	With recall
Distinct priority profiles	11	37
Distinct top-10 lists	16	28
Mean top-10 overlap between patients with the same cuisine	0.95	0.93
Top 10 meeting all three criteria	0.837	0.831

Default ranking, including acceptability. Without acceptability the recall gives 33 distinct lists: well-rated recipes are shared between patients.

Table 4. Recommendations move in the direction of each patient's priorities (mean of the top 10).

Priority	Level: high	Level: medium	Level: low
Saturated fat (% of energy)	3.20 (n=175)	3.66 (n=86)	3.63 (n=39)
Total sugars (% of energy)	4.19 (n=53)	4.31 (n=190)	4.66 (n=57)
Fibre (g per 1,000 kcal)	35.45 (n=255)	34.86 (n=37)	38.22 (n=8)

Patients with a high saturated-fat or sugar priority receive recipes lower in that nutrient. Fibre shows no gradient because almost every recommended recipe already exceeds the cap of the fibre margin (28 g per 1,000 kcal).

4. Scenario Engine, Layer 5 (§8)

A scenario is a set of recipes that replace the patient's usual meals. For each of the patient's criteria the engine compares the scenario with the patient's usual intake from the recall (usual meal energy is a third of daily energy) and reports whether it improves, worsens or does not change meaningfully. Each criterion carries the evidence rows for its pathway, and the scenario's label is the strongest label among the criteria that improve. The Layer 2 model is not queried for scenarios, because it was not trained on interventions, so no CAP or probability is ever predicted for one.

- **Energy:** calorie restriction improved steatosis (HAI2022-CR, MODERATE). Weight loss of 5% or more (VG2015-WL5, STRONG) is shown only as conditional: "if sustained lower energy leads to ≥ 5% weight loss".
- **Saturated fat:** saturated-fat overfeeding raised liver fat more than unsaturated fat (LUU2018-SAT, MODERATE), with the EASL guideline.
- **Sugars:** sugar harmed the liver when it added surplus energy (CHI2014-HYPER, MODERATE) but not when exchanged isocalorically (CHI2014-ISO, WEAK). A sugar reduction is graded MODERATE only when the scenario also lowers energy.
- **Fibre,** as a marker of a Mediterranean-style diet: the Mediterranean diet reduced MRI-PDFF (STE2026-MEDDIET, MODERATE). Because one recipe is not a dietary pattern, every scenario is marked as indirect evidence and labelled "Simulated projection, not a clinical forecast".

Table 5. Worked example, SEQN 110006 (predicted CAP 249.3 dB/m, grade S1; P(CAP ≥ 248) 50%; European cuisine).

Scenario	Criterion	Usual	Scenario	Direction	Evidence
Scenario A	Energy per serving (kcal)	398.3	318.9	improves	MODERATE
Scenario A	Fibre (g per 1,000 kcal)	7.3	39.6	improves	MODERATE
Scenario A	Saturated fat (% of energy)	8.4	1.9	improves	MODERATE
Scenario A	Total sugars (% of energy)	27.4	0.3	improves	MODERATE
Scenario B	Energy per serving (kcal)	398.3	393.2	no meaningful change	–
Scenario B	Fibre (g per 1,000 kcal)	7.3	17.5	improves	MODERATE
Scenario B	Saturated fat (% of energy)	8.4	7.6	no meaningful change	–
Scenario B	Total sugars (% of energy)	27.4	7.2	improves	WEAK / INCONCLUSIVE

Scenario A (Broccoli and Garlic Pasta; Southern Lima Butter Beans With Ham Bits; 5 Ingredient Soup): improvement, MODERATE, large shift from the usual diet. Scenario B (Socca (Provencal Savory Chickpea Pancake) - Gluten-Free; Lemon-Shrimp Pasta; Sugar - Browned Potatoes): improvement, MODERATE, moderate shift. In B the sugar reduction is WEAK because energy does not fall.

For the Layer 6 visualisation the engine also returns a timeline (today, 3 months, 6 months, 1 year) with the current grade and, for each horizon, only a direction and its evidence label.

5. Evidence Table Additions (§17.13)

Every value below was checked against the PubMed abstract. A new label, OBSERVATIONAL ASSOCIATION, keeps the coffee findings apart from intervention evidence; an automated test confirms that no priority rule cites them.

Table 6. Rows added to the evidence table (now 23 rows).

ID	Finding	Label
LIU2015-COF-FIB	Coffee: lower odds of advanced fibrosis (OR 0.73; 95% CI 0.58–0.92) and cirrhosis (OR 0.61); 16 studies, mixed liver diseases	OBSERVATIONAL ASSOCIATION
EBA2021-COF-FIB	Coffee in NAFLD: lower odds of significant fibrosis (RR 0.65; 95% CI 0.54–0.78); 5 studies, 4,303 participants	OBSERVATIONAL ASSOCIATION
EBA2021-COF-INC	Coffee: no association with NAFLD incidence (RR 0.88; 0.63–1.25) or prevalence (RR 0.88; 0.76–1.02)	OBSERVATIONAL ASSOCIATION
STE2026-FAST	Fasting: ALT −12.47 IU/L (6 RCTs) and liver stiffness −0.24 kPa (4 RCTs)	MODERATE
STE2026-LCHF	Low-carbohydrate high-fat / ketogenic diets: no significant change in ALT (6 RCTs)	WEAK / INCONCLUSIVE
STE2026-OMEGA3	Omega-3 supplements: no significant change in ALT (4 RCTs)	WEAK / INCONCLUSIVE

Sources: Liu et al., PLoS One 2015 (PMID 26556483); Ebadi et al., Nutrients 2021 (PMID 34578919); Stern et al., Eur J Nutr 2026 (PMID 41689666).

6. Acceptability, Diversity and Seven-Day Meal Plans (§17.6, §17.12)

Acceptability is the RecipeDB rating shrunk towards the corpus mean with a pseudo-count of 20, because ratings cluster at five stars (the 25th percentile is 5.0); it matters through the number of ratings. At weight 0.1 it improved every measure, so it is now part of the default ranking. **Diversity** re-ranking (maximal marginal relevance over ingredient sets, ignoring seasonings and pantry staples) added little, because the default top 10 is already varied, so it remains an option.

Table 7. Ranking variants on 300 held-out patients (24-hour recall used).

Method	Health	Cuisine	Feasible	All three	Diversity	Acceptability
MC, health-gated (default)	1.000	0.998	0.833	0.831	0.947	88.6
MC, without acceptability	1.000	0.995	0.806	0.802	0.946	83.9
MC + diversity re-ranking (MMR)	1.000	0.990	0.816	0.807	0.957	88.6
MC, weighted only	0.742	1.000	0.968	0.730	0.952	91.1
Health score only	1.000	0.057	0.400	0.017	0.966	98.6
Nutrition only (non-personalised)	1.000	0.057	0.400	0.017	0.966	98.6
Random	0.057	0.053	0.596	0.003	0.962	80.9

MC: multi-criteria; the default includes acceptability (weights 0.6, 0.2, 0.2, 0.1). Shares of each top 10 meeting each criterion. Feasible: at most 60 minutes, 15 ingredients and 12 steps. Diversity: mean pairwise 1 − Jaccard of ingredient

sets. Cuisine preferences are simulated because NHANES does not record them. Only the health-gated variants always meet every health reference. Weight sensitivity: median top-10 overlap with the default 0.79 over 36 weight combinations.

Meal plans give two main meals a day for seven days from recipes that meet every reference, practical recipes first. Each plan is checked for: every meal and every day meeting the references, a recipe at most twice and never on the same or consecutive days, different protein families within a day and at most three per week, and the requested cuisine. A patient may not have the ingredients, tools or taste for another cuisine, so the week is filled in three steps and stops at the first that completes it: (1) the patient's own cuisine only; (2) a favourite recipe may repeat; (3) only then, recipes from the most similar cuisines, each labelled as borrowed. Similarity is the cosine of the cuisines' ingredient profiles across the corpus, at least 0.8: American → Mexican (0.96); Canadian → Australian (0.90), European (0.90).

Table 8. Seven-day plans for the 300 patients.

Check	Share of patients
Every meal meets the references	1.000
Every day meets the references	1.000
Repeats within limits; protein varied	1.000
Complete with own cuisine only (step 1)	0.583
Complete by step 2 (a favourite may repeat)	0.810
Complete by step 3 (similar cuisine allowed)	1.000

Asian, European and Mexican plans always complete at step 1; American plans complete at step 2 without borrowing; Canadian plans need step 3 and borrow 0.95 meals on average. Plans used 5.8 protein families and repeated 1.0 recipes on average; 81% of plan meals are within the practical limits.

Table 9. Example seven-day plan for SEQN 110006, American cuisine (completed at step 2).

Day	Meal 1	Meal 2
Day 1	5 Ingredient Soup (legumes)	Chipotle Pinto Beans (pork) [repeat]
Day 2	Veggie Tex-Mex Tortilla Wraps (legumes)	Huevos Rancheros Tostada (cheese) [repeat]
Day 3	Black Bean and Couscous Salad (legumes)	Fiesta Taco Casserole (cheese)
Day 4	Mexican Chicken Soup With Cilantro-Chile Cream (chicken)	Mexican Charro Beans (pork)
Day 5	Southwest Chicken Black Bean Soup (chicken)	Ground Turkey Chili (turkey)
Day 6	Slow Cooker Latin Chicken W/ Sweet Potatoes and Black Beans (chicken)	Crock Pot Posole (plant-based)
Day 7	Chipotle Pinto Beans (pork) [repeat]	Huevos Rancheros Tostada (cheese) [repeat]

Protein family in brackets. Two favourites appear twice, never on consecutive days.

7. Data Access Findings

- **RecipeDB.** The API available to the project reports 20,000 recipes in total, and filtering by cuisine returns 19 Indian recipes — the same 20,000 already in the pipeline. Of the 9,844 candidate meals, only 565 meet every reference value and 324 of those have no

specific cuisine label. Since the patient data are from the US population, US cuisines are the focus and the regional Indian expansion is now future work (§17.14).

- **FPED.** USDA provides FPED only up to 2017–March 2020, so added sugars and HEI-2020 are missing for the 2021–2023 cycle; sugar priorities for those patients are not adjusted.
- **DR1IFF** (individual foods) turned out not to be needed: FPED provides participant-level totals.

8. Limitations

- One 24-hour recall describes one day, not habitual intake; it is used only to move a priority by one level.
- Cuisine preferences in the evaluation are simulated; the pilot evaluation is the only test with real preferences.
- Scenario directions rest on evidence about dietary patterns sustained for weeks, not single recipes.
- Recipe sugars are total sugars, while the patient's sugar priority uses added sugars, matching the available data on each side.
- The protein family is a keyword heuristic on the first protein ingredient and is sometimes wrong (for example, a chicken and feta pasta is labelled cheese).
- Repeating a favourite and borrowing from a similar cuisine are design choices not yet tested with users; the pilot evaluation should ask whether people would accept them.

9. Decisions Taken This Week

These decisions shape the results above; they are listed so that they can be confirmed or revisited.

1. **Personalisation over the headline score.** The ranking now lets priority levels count inside the health gate, accepting 0.802 instead of 0.899 for the top 10 meeting all three criteria, because the 0.899 came from an engine that gave almost every patient the same list. A digital twin that ignores the patient is not personalised.
2. **Acceptability in the default ranking** (weight 0.1): all three criteria 0.802 → 0.831, and recipes many people have rated well are preferred.

3. **US cuisines first; Indian regional cuisines are future work.** The cohort is the US population and the available RecipeDB holds 19 Indian recipes.
4. **Cuisine mixing only as a last resort.** Meal plans use the patient's own cuisine, then repeat a favourite, and only then borrow from the most similar cuisine by ingredients, labelling those meals.

10. Next Steps (Meeting 10, Friday 30 October)

- Recruit the 5–10 pilot evaluators (carried over from meeting 9).
- Fibrosis risk (FIB-4) validated against measured liver stiffness, with the referral pathway (§17.2); MASLD/MetALD labelling and safety guardrails (§17.3).
- Per-patient SHAP and conformal CAP intervals (§17.4); the causal knowledge graph (§5.3).
- Knowledge-graph-traced recipe explanations (§7.3), grounded plain-language summaries (§17.7), and the improved cuisine and feasibility scores (§7.1).
- Download the NHANES III ultrasound and Linked Mortality Files and map variables (§17.9).

Reproducing these results: python -m src.build_habitual_diet --download; python -m src.evaluate_ranking (and --no-diet for the comparison run); python src/demo.py --seqn 110006 --cuisine European --plan.